

Executive Summary

This notebook presents a multi-task model aimed at detecting mental health conditions from user inputs and output empathetic responses. It uses two NLP models: XLM-RoBERTa for sentiment classification and FLAN-T5 (Large) for supportive message generation. The model is designed for applications in mental health detection on social media, automated mental health support platforms, and psychological counselling assistance.

The training dataset was sourced from Kaggle, which contain user texts labelled with seven different health categories: Normal, Depression, Suicidal, Anxiety, Bipolar, Stress, and Personality Disorder. The data was pre-processed and then split into training, validation, and test sets. User texts were converted into tokens using Hugging Face's XLM-RoBERTa tokenizer, a multilingual pre-trained model supporting 100+ languages. To address class imbalance, weighted loss function was employed, using a custom WeightedTrainer, which ensured minority classes like "Personality disorder" received fair representation during loss computation. Hugging Face's TrainingArguments class was used in training which allowed specification of key hyperparameters and management of the overall training workflow.

The fine-tuned model achieved an overall accuracy of 78% and a macro F1-score of 75%, with notably stronger performance in 'Normal', 'Anxiety', and 'Bipolar'. Some vague inputs were harder to classify accurately which suggested further improvements using prompt tuning and more domain-specific data. The inclusion of topic modelling in the future could allow us to identify feature clusters explaining the underlying cause of emotional distress.

Overall, the model proves to be an efficient and compassionate AI system for mental health monitoring, allowing specialists to assess the need for intervention. FLAN-T5, a pretrained text-to-text generation model, was integrated to the model, enabling generation of comforting messages tailored to the detected mental health condition for real-time support. Limitation such as the ability to generate responses only in English, can be addressed in future enhancements.